Software Quality Evaluation Report

Section 1: Cover Information

App Names:	App 1: Spotify | App 2: YouTube Music
Version Numbers:	Spotify (8.9.94.614) | YT Music (7.02.52)
Platform/OS:	Android 14 (Samsung Galaxy)
Evaluator:	Haseeb Rauf
Roll No:	FA23-BSE-028
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Section 2: App Overviews

Spotify

Spotify is a leading digital music service providing access to millions of songs and podcasts. Its primary purpose is to deliver a personalized audio streaming experience through high-quality discovery algorithms. Target users range from casual listeners to music enthusiasts. Main features include Spotify Connect, offline playback, and collaborative playlists.

YouTube Music

YouTube Music integrates the vast video library of YouTube into a music-specific streaming interface. Its primary purpose is to merge video performances with audio streaming. Target users are often Google ecosystem participants. While both apps share the category of music streaming, YouTube Music is uniquely positioned to offer user-generated and live video content.

Section 3: Evaluation Methodology

Testing utilized a 1–5 Likert scale where 5 indicates excellence. Empirical data was gathered on a Samsung Galaxy (Android 13) over 120 minutes per app. Tools used included a digital stopwatch for performance timing, system settings for resource monitoring, and TalkBack for accessibility testing.

Section 4: Completed Checklists

Below is the summary of the ratings gathered across the 8 ISO 25010 characteristics.

Characteristic	Weight	Spotify	YT Music
Functional Suitability	15%	5	4
Performance Efficiency	12%	4	4
Compatibility	10%	5	5
Usability	15%	4	4
Reliability	15%	4	4
Security	15%	4	4
Maintainability	8%	4	5
Portability	10%	5	5

Section 5: Quality Profile Chart

Section 6: Analysis

Following an extensive empirical evaluation, Spotify achieved a weighted quality score of 4.35, while YouTube Music followed closely with a score of 4.27. This results in a technical margin of 0.08 points. While both applications represent the upper echelon of the music streaming market, the margin indicates that Spotify maintains a slight edge in overall software quality. This lead is primarily attributed to Spotify's maturity as a standalone audio platform, which allows it to focus exclusively on audio-centric functional suitability without the architectural complexity of managing a parent video-sharing platform's data structures.

The most significant quality gap was identified in Functional Suitability, where Spotify secured a 5 compared to YouTube Music's 4. In practice, this gap manifests in library management; Spotify's 'Liked Songs' and folder structures feel more intentional and less cluttered. In contrast, YouTube Music often suffers from 'Algorithm Pollution,' where user interactions with non-music videos inadvertently clutter the music-specific 'Your Likes' playlist. The second gap appeared in Maintainability, where YouTube Music actually outperformed Spotify (5 vs 4). Due to Google's unified backend, YouTube Music updates are deployed with extreme frequency and high reliability. The final gap was noted in Usability; Spotify's visual hierarchy on the 'Now Playing' screen is more conducive to high-speed interactions.

A critical trade-off was observed between Security and Maintainability. Spotify maintains high security but relies on third-party authentication (OAuth) via Facebook or Google. YouTube Music benefits from native Google Account security infrastructure. However, this high maintainability comes at the cost of Ecosystem Lock-in. For the 'Power User'

prioritizing library control, Spotify is recommended. For the 'Ecosystem User' already invested in Google services, YouTube Music is the logical choice despite the 0.08 margin.

Section 7: Recommendations

- Spotify should implement a 'Low Power Mode' in order to improve Performance Efficiency from its current rating of 4 to a target of 5. Currently, background RAM usage exceeds 500MB. This change would extend device battery life during long listening sessions.
- Spotify should integrate native 2FA in order to improve Security from its current rating of 4 to a target of 5. Currently, it relies on external providers. This change would significantly harden user accounts.
- YouTube Music should separate Video and Music likes in order to improve Functional Suitability from its current rating of 4 to a target of 5. Currently, library management is cluttered. This change would provide a cleaner experience.
- YouTube Music should implement background token refresh in order to improve Reliability from its current rating of 4 to a target of 5. Currently, offline downloads fail if re-verification is skipped. This change ensures uninterrupted music.